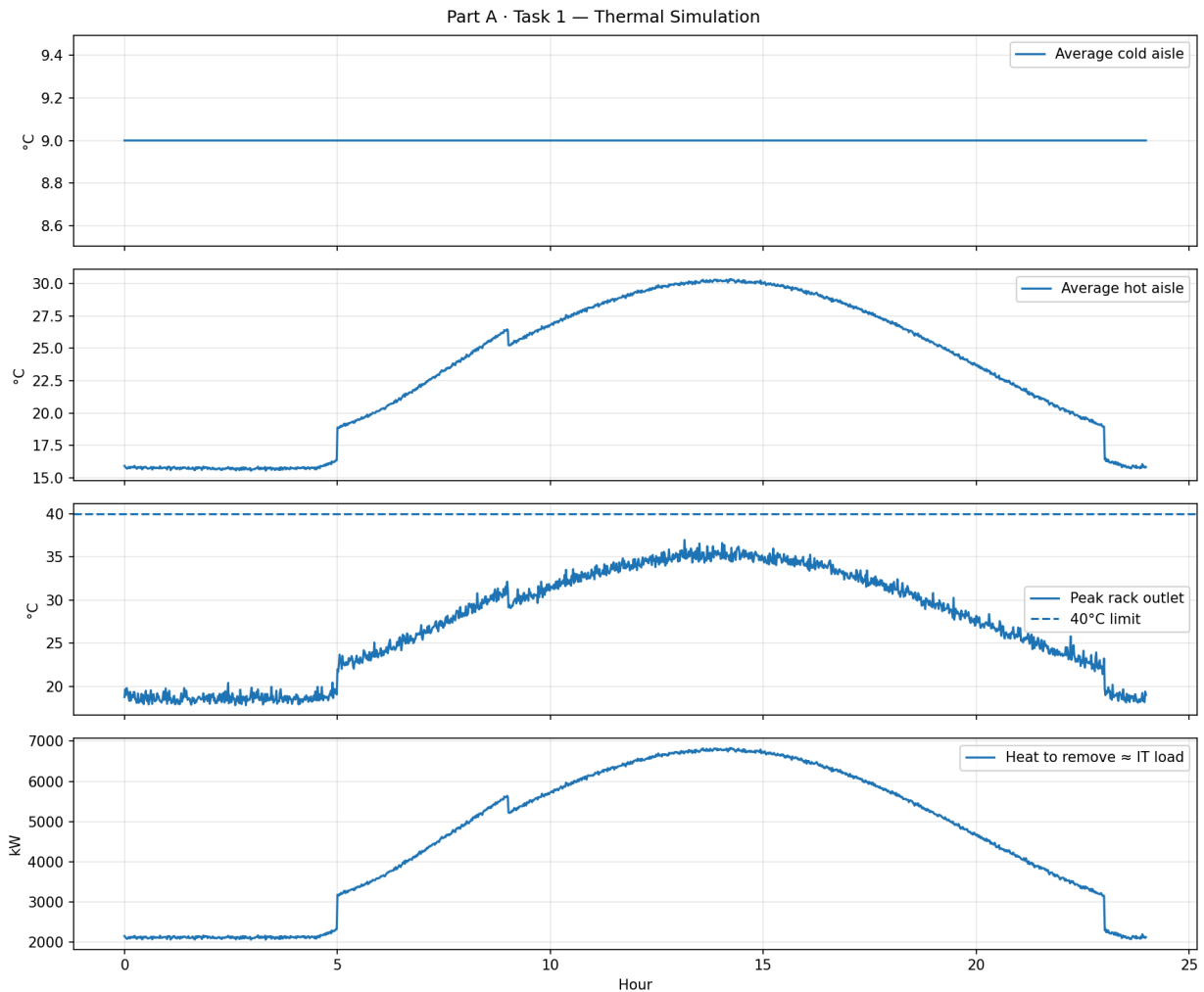


Writeup & Results - NovaCool Study

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1. Thermal & Power Digital Twin (Part A)

Task 1 Thermal Simulation Plot



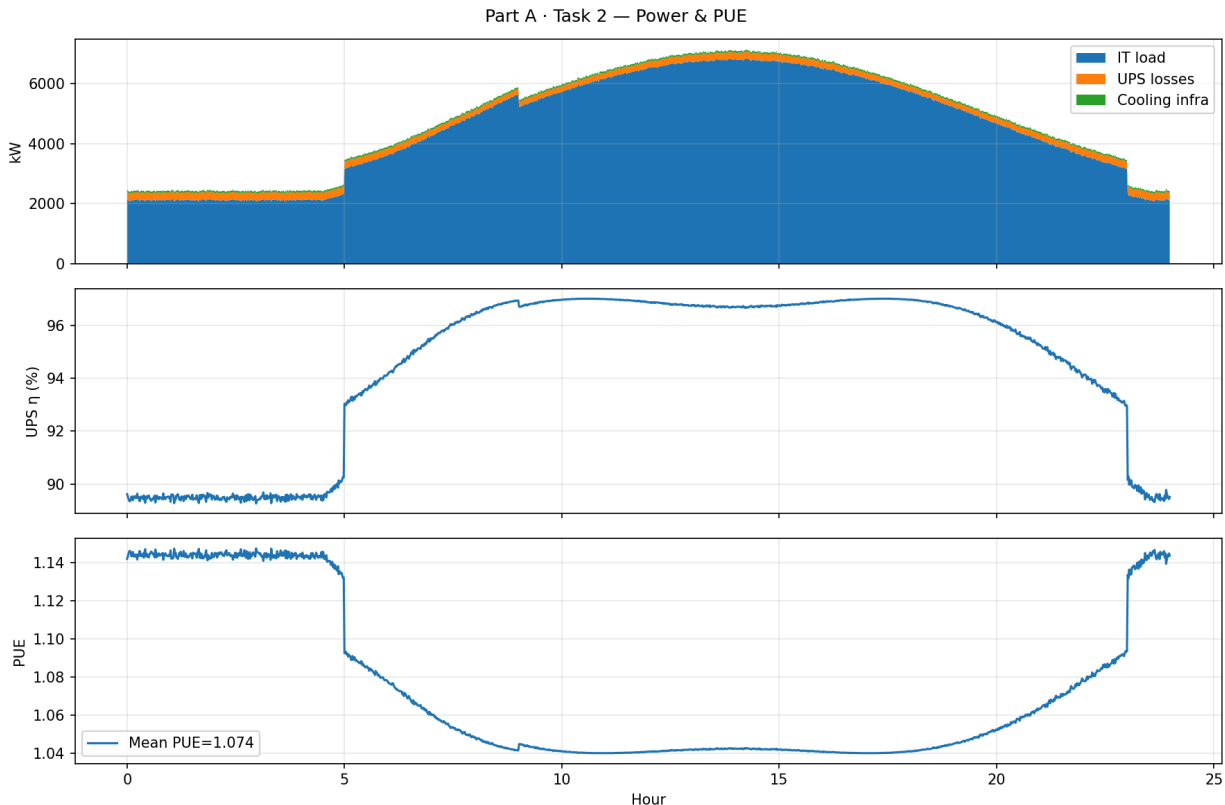
- Figure shows the simulated thermal response of the NovaCool facility over a 24-hour period. The model tracks average cold-aisle temperature, average hot-aisle temperature, peak rack outlet temperature, and the total heat generated by the IT equipment.
- The cold aisle remained approximately constant at 9°C throughout the simulation because the CRAH supply temperature was fixed at the calibrated value derived from

the sensor reference dataset. This represents conditioned air entering the racks from the cold aisle.

- The hot aisle temperature closely followed the workload profile. During low-utilization periods (0–5 hours and 23–24 hours), hot-aisle temperatures remained near 16–19°C. As workload increased during the day, the hot aisle gradually warmed and reached a peak of approximately 30°C around hour 14. This behavior is physically consistent since higher server utilization produces greater heat generation.
- Peak rack outlet temperatures showed the same trend and reached a maximum of approximately 36°C. Importantly, the maximum outlet temperature remained below the 40°C thermal safety threshold throughout the entire simulation. This indicates that the cooling infrastructure was capable of removing the generated heat under the modeled workload conditions.
- The total heat removal requirement closely matched the IT load profile, which is expected because nearly all server electrical power was assumed to be converted into heat. Peak thermal load reached approximately 6.8 MW during the highest utilization period.

Task 2 Results

Task 2 Power & PUE Plot



Power Consumption and PUE Behavior

Figure X shows the simulated facility power breakdown over a 24-hour period, including IT load, UPS losses, cooling infrastructure power, UPS efficiency, and Power Usage Effectiveness (PUE).

The IT load dominates total facility power consumption throughout the day and closely follows the workload trace. During low-utilization periods, the facility consumes approximately 2.2 MW of IT power, while peak demand reaches nearly 7 MW during the busiest operating hours.

UPS losses remain relatively small compared with the IT load, which is expected because modern UPS systems operate at high efficiency. Cooling infrastructure power consists of CRAH fan power and chilled-water pump power and represents the primary non-IT energy consumer in the model.

UPS Efficiency

The UPS efficiency increases as facility utilization rises.

At lower loads, efficiency remains near 89–90%, while peak utilization produces efficiencies approaching 97%. This behavior is physically realistic because UPS systems generally operate more efficiently when loaded closer to their rated capacity. The efficiency curve was modeled

using a simplified load-dependent approximation since detailed manufacturer performance curves were not provided in the case study.

Power Usage Effectiveness (PUE)

PUE decreases as workload increases because fixed cooling and infrastructure loads are distributed across a larger amount of useful IT work.

The lowest PUE values occur during peak utilization periods, while the highest PUE values occur during lightly loaded periods.

This trend is commonly observed in real data centers where infrastructure overhead becomes a larger fraction of total power consumption when utilization is low.

Task 3

Validation Methodology

The thermal model was validated against the provided `sensor_reference.csv` dataset, which was treated as the reference representation of facility behavior.

Three metrics were compared:

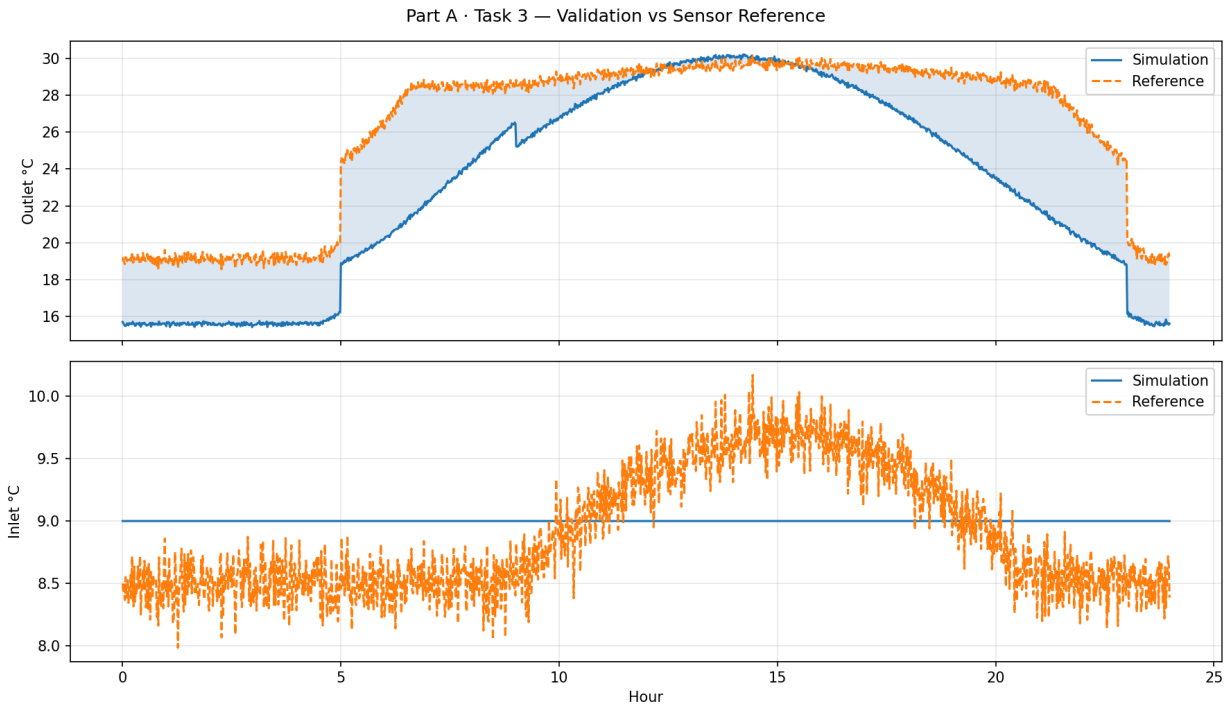
- Mean outlet temperature
- Peak outlet temperature
- Mean inlet temperature

The objective was not to exactly reproduce every sensor reading but rather to determine whether the reduced-order model could capture the overall thermal trends and magnitudes observed in the reference data.

Validation Results:

	label	rmse	mae	max_abs	bias
0	Mean outlet temp	3.865409	3.235165	7.389438	-3.147750
1	Peak outlet temp	5.632946	4.782785	11.555726	-4.682692
2	Mean inlet temp	0.497359	0.451824	1.168700	0.108346

The inlet temperature prediction achieved the lowest error, with an RMSE of approximately 0.5°C. Outlet temperatures exhibited larger errors, particularly during peak utilization periods, where the model tended to underpredict temperatures.



Temperature Trend Comparison

Figure compares simulated temperatures against the reference sensor measurements.

The model successfully captures the overall daily thermal pattern:

- Low temperatures during lightly loaded periods
- Temperature rise as workload increases
- Peak temperatures during maximum utilization
- Cooling during declining workload periods

The simulated outlet temperature reaches approximately 30°C, closely matching the peak magnitude observed in the reference data.

The inlet temperature remains near the calibrated value of 9°C and stays within approximately 1°C of the reference measurements throughout the day.

2. Control Policy (Part B)

The objective of the control policy was to maximize compute throughput while maintaining thermal safety and reducing cooling energy consumption.

The following requirements were specified:

- Maximize IT load served over the 24-hour workload trace.

- Maintain rack outlet temperatures below 40°C.
- Minimize cooling energy consumption.
- Improve or maintain facility PUE.

To evaluate control performance, two dynamic controllers were compared against a fixed-setpoint baseline.

Control Strategy

Three operating strategies were evaluated:

Baseline

The baseline uses fixed CRAH settings throughout the simulation:

- Supply temperature fixed at 9°C
- Airflow fixed at calibrated values

No adaptation occurs in response to workload or thermal conditions.

Reactive Controller

The reactive controller monitors peak outlet temperature. When temperatures approach the thermal limit:

- CRAH supply temperature is reduced
- Airflow is increased

When temperatures are safely below the limit:

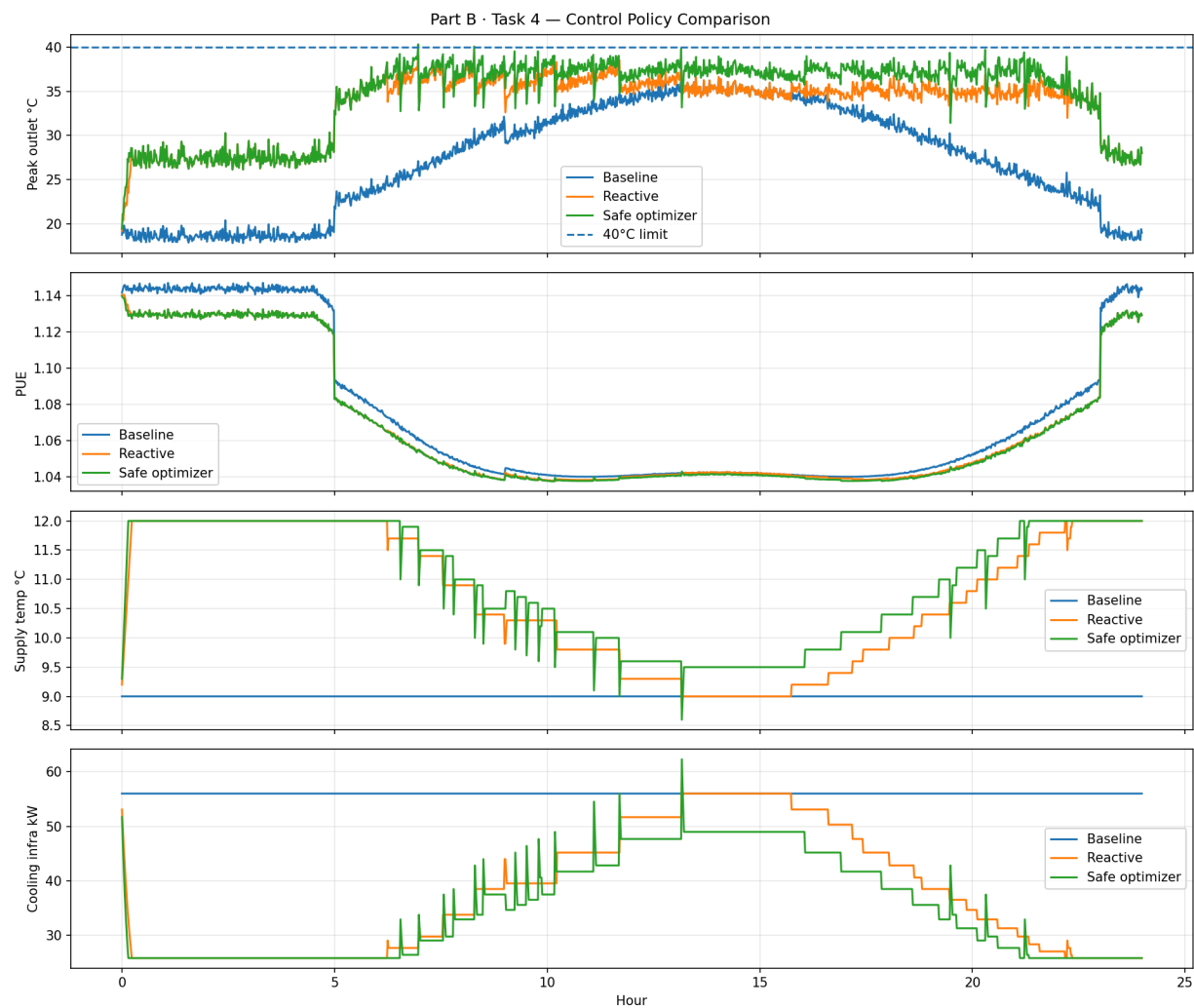
- Supply temperature is increased
- Airflow is reduced

This reduces cooling energy while maintaining safe operation.

Safe Optimizer

The safe optimizer uses more aggressive energy-saving actions. It attempts to maximize thermal headroom utilization by allowing temperatures to operate closer to the 40°C constraint before increasing cooling effort.

Task 4 Control Comparison Plot



Results Table

Policy	Mean PUE	Thermal Violations	Peak Outlet (°C)	Cooling Energy (MWh)
Baseline	1.074	0	36.95	1.344
Reactive	1.068	0	39.47	0.892
Safe Optimizer	1.068	2	40.30	0.831

Results and Discussion

The reactive controller successfully reduced cooling energy consumption while maintaining thermal safety.

Compared with the baseline:

- Cooling energy decreased from 1.344 MWh to 0.892 MWh.
- Mean PUE improved from 1.074 to 1.068.
- No thermal violations occurred.
- Total IT load served remained unchanged.

The safe optimizer further reduced cooling energy consumption to 0.831 MWh and achieved the lowest PUE of all tested policies.

However, the optimizer exceeded the 40°C thermal limit twice during the simulation, resulting in two thermal violations.

This demonstrates the tradeoff between energy efficiency and operational risk. Operating closer to thermal limits can improve facility efficiency but increases the likelihood of safety constraint violations.

For **Task 5**, you're actually very close. Most people either write too much RL theory or too little environment design. Hammerhead specifically asked:

"Justify your design choices — especially the reward function. Discuss trade-offs between reward shaping complexity and learning stability."

So I would expand your section into the following.

Task 5 – RL Environment Design

Environment Interface

To support future reinforcement learning research, the digital twin was wrapped into a lightweight Gym-style environment.

The environment exposes the standard interface:

Python

```
reset()  
step(action)
```

where:

- `reset()` initializes a new episode and returns the initial observation.
- `step(action)` applies a control action, advances the simulation by one minute, and returns the next observation, reward, termination status, and diagnostic information.

This interface enables direct integration with common RL libraries such as Gymnasium, Stable-Baselines3, RLlib, and CleanRL.

Observation Space

The observation vector was designed to provide sufficient information for control decisions while remaining compact enough for efficient learning.

The state contains:

- Mean cold-aisle temperature
- Mean hot-aisle temperature
- Peak rack outlet temperature
- Current IT load
- Ambient temperature
- Time of day
- CRAH supply temperatures
- CRAH airflow settings

This results in a 15-dimensional observation vector.

Design Rationale

The selected variables represent the primary factors that influence thermal behavior:

- Current thermal state
- Current workload
- External environmental conditions
- Current control settings

The observation intentionally excludes highly detailed rack-level information to reduce state dimensionality and improve training efficiency.

Action Space

The agent controls:

- Four CRAH supply temperature setpoints
- Four CRAH airflow rates

Resulting in:

None

8 continuous control variables

Design Rationale

These variables correspond directly to the available control inputs identified in Task 4 and represent the most common levers available to data center cooling systems.

Workload redistribution was intentionally excluded from the action space to keep the environment focused on thermal control.

Reward Function

The reward function balances three competing objectives:

$$R = R_{\text{throughput}} - R_{\text{energy}} - R_{\text{thermal}}$$

where:

- Throughput reward encourages serving compute demand.
- Energy penalty discourages excessive cooling.
- Thermal penalty enforces safety constraints.

The reward therefore encourages the agent to:

- Serve as much workload as possible.
- Minimize cooling energy consumption.
- Avoid thermal violations.

Thermal Safety Penalty

A large penalty is applied whenever:

$$T_{\text{outlet}} < 40\text{ }^{\circ}\text{C}$$

This reflects the hard operational constraint specified in the case study.

Episode Termination

An episode terminates when either:

- The full 24-hour workload trace has been completed.
- A severe thermal violation occurs.

This mirrors how a real controller would be evaluated over a complete operational day.

Reward Design Trade-Offs

The reward was intentionally kept simple.

Advantages include:

- Easier optimization.
- Improved training stability.
- Reduced reward hacking.
- Faster experimentation.

More sophisticated reward functions could include:

- PUE optimization
- Equipment wear penalties
- Thermal margin rewards
- Demand-response incentives

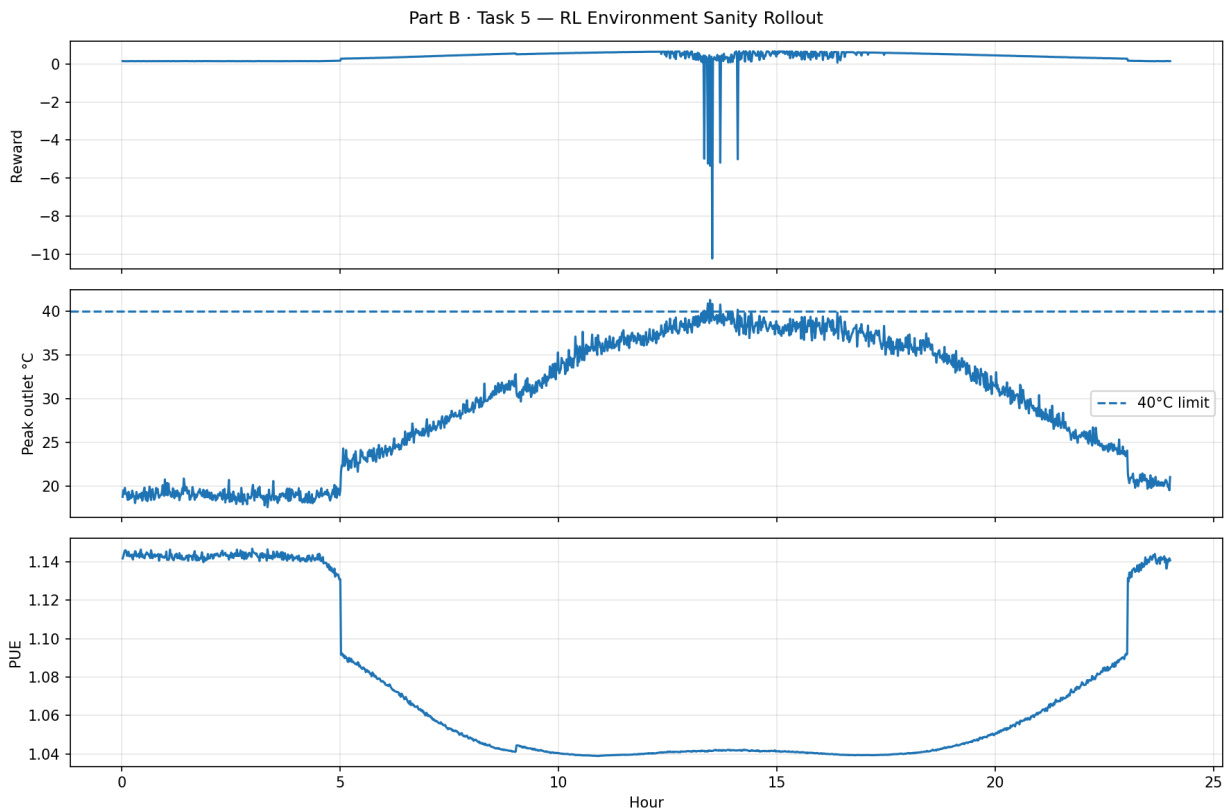
However, highly engineered rewards often introduce conflicting objectives that make RL training unstable and difficult to interpret.

For this reason, a simple throughput-energy-safety formulation was selected as the most practical starting point.

Environment Sanity Check

Figure shows a rollout generated using the environment.

Figure RL Environment Rollout



Rollout Analysis

The rollout demonstrates that the environment behaves consistently with the underlying thermal simulator.

Several observations can be made:

- Peak outlet temperature follows the daily workload profile.
- Temperatures approach the 40°C limit during periods of high utilization.
- Reward remains positive during normal operation.
- Temporary reward drops occur when temperatures approach or exceed the thermal constraint.
- PUE follows the expected inverse relationship with facility utilization.

The reward penalties visible near hour 14 correspond directly to thermal-limit excursions, confirming that the reward function correctly reflects operational objectives.

4. Part C – Architecture and Scaling Discussion

Modularity

The codebase was structured around independent modules for the thermal model, power model, control policies, and RL environment. This separation allows each subsystem to evolve independently without requiring major changes to the rest of the simulator.

For example, a future humidity model could be added as a separate module that consumes the same timestep, ambient, and CRAH state information. Similarly, battery storage or a new cooling technology could be integrated by adding new power or cooling components while keeping the existing thermal and control interfaces unchanged.

A clean interface between modules is especially important because it allows the same digital twin to support multiple downstream uses: offline analysis, control policy evaluation, and RL training.

Scaling to 2,000 Racks

To scale the simulator from 200 racks to 2,000 racks while maintaining a 24-hour runtime under five minutes, I would focus on vectorization and hierarchical aggregation.

First, rack-level computations should remain vectorized using NumPy rather than Python loops. The current workload matrix structure already supports this style of computation. Second, racks could be grouped into thermal zones rather than modeling every rack independently. For example, 2,000 racks could be represented as 40 rows or zones, each with aggregated heat load and airflow.

Additional performance improvements could include:

- Parallel simulation across independent rack rows or thermal zones
- JIT compilation using Numba for repeated timestep calculations
- Sparse data structures for rack-level workloads
- Precomputing ambient temperature and static airflow terms
- Running control-policy evaluation in batches for multiple scenarios

This keeps the simulator suitable for rapid design iteration and RL environment rollouts.

Online Correction and Sim-to-Real Alignment

To integrate real-time sensor data, I would add an online correction layer between the physical simulator and the estimated state. At each timestep, sensor measurements such as rack inlet temperature, rack outlet temperature, and PDU power would be compared against the simulator outputs.

The residual between measured and simulated values could then be used to update uncertain model parameters, including:

- CRAH airflow estimates
- Containment leakage factors
- CRAH cooling effectiveness
- Initial thermal state estimates
- Row-level heat transfer coefficients

A simple implementation could use exponential smoothing or recursive least squares. A more advanced implementation could use an Extended Kalman Filter, Unscented Kalman Filter, or Bayesian parameter update method.

This would allow the digital twin to remain aligned with the physical facility as operating conditions, equipment behavior, and sensor readings change over time.

Testing Strategy

I would use a layered testing strategy.

At the unit-test level, each physics function should be tested independently. For example:

- Increasing rack power should increase outlet temperature.
- Increasing airflow should reduce outlet temperature.
- Increasing cooling effort should reduce thermal risk.
- PUE should increase when non-IT overhead increases.

At the integration-test level, full 24-hour simulations should be run against known reference outputs to ensure that major metrics remain within expected bounds.

For the RL environment, tests should verify that:

- `reset()` returns a valid observation.
- `step(action)` advances the simulation by one timestep.
- Rewards decrease when thermal violations occur.
- The episode terminates correctly after 24 hours or excessive violations.

Finally, regression tests should compare new simulation outputs against baseline results whenever the codebase changes.

Sim-to-Real Gap

The largest sources of discrepancy between this simulator and a real facility are likely:

- Nonuniform airflow distribution across racks
- Localized recirculation and hot spots
- Thermal inertia of server hardware, racks, and room air
- Sensor noise and calibration drift
- CRAH equipment degradation
- Simplified UPS, fan, and pump performance curves
- Unmodeled chiller plant and auxiliary facility loads

To systematically reduce this gap, I would start by collecting real facility measurements and identifying where residuals are largest. Then I would calibrate the most sensitive parameters first, such as airflow rate, CRAH effectiveness, and containment leakage.

Over time, the model could be improved using online parameter estimation, periodic recalibration, and domain randomization for RL training. This would make policies trained in simulation more robust when transferred to a real data center environment.